UNIT I : Introduction to Heat Transfer

Subject: Heat Transfer (23A03601T) Topics: Modes, conduction eqn, slabs, cylinders, critical radius, fins, transient conduction

1.1 What is Heat Transfer?

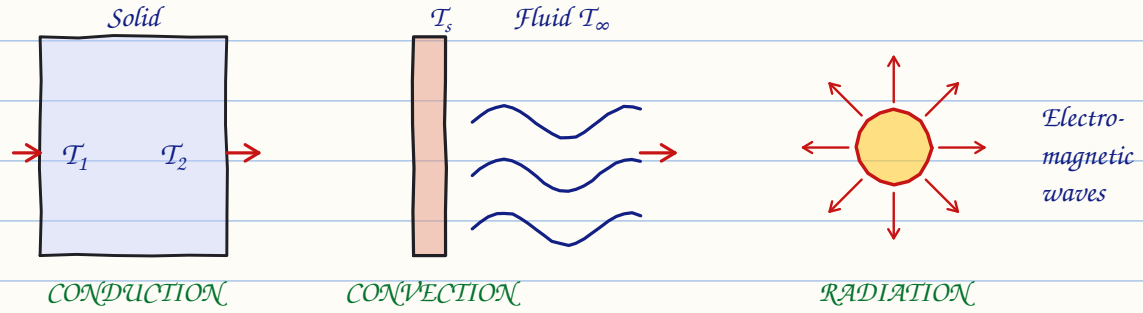
Heat transfer is the study of energy in transit because of a temperature difference. Thermodynamics tells us how much energy is transferred (end states, equilibrium); heat transfer tells us how fast (rate) and by what mechanism.

→ Driving potential : temperature difference ΔT (like voltage in electricity)

→ Heat always flows from a high temperature region to a low temperature region (2nd law)

→ Unit of heat rate Q : Watt ($W = J/s$); heat flux $q = Q/A$: W/m^2

1.2 Basic Modes of Heat Transfer



Mode	Rate equation	Medium	Key property
Conduction	$Q = -kA\, dT/dx$	Solid / fluid at rest	k (W/mK)
Convection	$Q = hA(T_s - T_\infty)$	Moving fluid	h (W/m^2K)
Radiation	$Q = \varepsilon\sigma A(T_1^4 - T_2^4)$	No medium (vacuum)	ε, $\sigma = 5.67\times10^{-8}$

1.3 Rate Equations

(a) Conduction – Fourier's law: rate of heat conduction is proportional to the area normal to flow and the temperature gradient.

$$Q = -kA\,(dT/dx)\ [W] \qquad \text{Fourier law}$$

→ k = thermal conductivity; -ve sign : heat flows in direction of falling T

→ k: metals (Cu 385, Al 205, Steel 45) > liquids (water 0.6) > gases (air 0.026) ; insulators 0.03–0.1

(b) Convection – Newton's law of cooling:

$$Q = h\,A\,(T_s - T_\infty)\ [W]$$ Newton law

→ h = convective heat transfer coefficient (W/m^2K); depends on velocity, fluid properties, geometry

→ Free convection $h \approx 2$–25 ; forced (air) 25–250 ; forced (water) 50–20,000 ; boiling/condensation 2500–100,000

(c) Radiation – Stefan–Boltzmann law:

$$E_b = \sigma\,T^4\ [W/m^2]$$

$$Q = \varepsilon\,\sigma\,A\,(T_1^4 - T_2^4)$$ Stefan–Boltzmann

→ $\sigma = 5.67\times10^{-8}\ W/m^2K^4$; T in Kelvin; ε = emissivity (0 to 1)

1.4 Thermal Diffusivity

$$\alpha = k / (\rho\,c_p)\ [m^2/s]$$

Ratio of heat conducted to heat stored. High α → heat diffuses quickly through the material (e.g., metals); low α → slow response.

Note: In real problems all three modes occur together (combined mode). e.g. heat loss from a hot pipe: conduction in the wall, convection to air and radiation to surroundings.

Thermal Conductivity – a closer look

k is a property : the heat conducted per unit area per unit time for unit temperature gradient. Mechanism: free-electron flow + lattice vibrations (metals), lattice vibrations (insulating solids), molecular collisions (fluids).

Material	k at 20°C (W/mK)	Material	k at 20°C (W/mK)
Copper	385	Water	0.60
Aluminium	205	Engine oil	0.145
Mild steel	45	Air	0.026
Brick (common)	0.72	Glass wool	0.04
Concrete	1.4	Asbestos sheet	0.16

→ k varies with temperature : $k = k_0(1 + \beta T)$. For pure metals k decreases with T; for gases k increases with T; for most insulators k increases with T

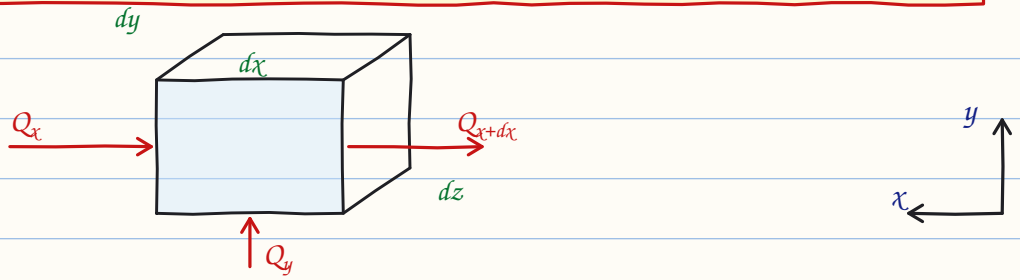
→ Impurities and alloying reduce k of metals (pure Cu 385 → brass 110); moisture in insulation increases k

→ Mean k is used if variation is small: $Q = k_m\,A\,\Delta T/L$ with $k_m = k_0[1 + \beta(T_1+T_2)/2]$

1.5 Generalised Heat Conduction Equation (Cartesian)

Consider an elemental volume $dx \cdot dy \cdot dz$ in a homogeneous solid with internal heat generation q_g (W/m^3). Energy balance on the element:

(Net heat conducted in) + (Heat generated) = (Rate of increase of internal energy)



Heat in along x : $Q_x = -k\,dy\,dz\,(\partial T/\partial x)$. Heat out : $Q_{x+dx} = Q_x + \partial/\partial x\,(Q_x)\,dx$. Net along $x = \partial/\partial x\,(k\,\partial T/\partial x)\,dx\,dy\,dz$. Similarly for y and z. Generation = $q_g\,dx\,dy\,dz$. Storage = $\rho\,c_p\,(\partial T/\partial t)\,dx\,dy\,dz$. Hence,

$$\partial/\partial x(k\,\partial T/\partial x) + \partial/\partial y(k\,\partial T/\partial y) + \partial/\partial z(k\,\partial T/\partial z) + q_g = \rho\,c_p\,\partial T/\partial t$$

1.6 Various Forms of the Conduction Equation

Constant k (homogeneous, isotropic): with $\alpha = k/\rho c_p$

$$\nabla^2 T + q_g/k = (1/\alpha)\,\partial T/\partial t$$ *Fourier–Biot eqn*

Condition	Equation	Name
No generation, unsteady	$\nabla^2 T = (1/\alpha)\,\partial T/\partial t$	Fourier eqn
Steady, with generation	$\nabla^2 T + q_g/k = 0$	Poisson eqn
Steady, no generation	$\nabla^2 T = 0$	Laplace eqn
1-D steady, no generation	$d^2T/dx^2 = 0$	Linear T profile

Cylindrical co-ordinates (r, φ, z):

$$(1/r)\,\partial/\partial r(\,r\,\partial T/\partial r\,) + (1/r^2)\,\partial^2 T/\partial \varphi^2 + \partial^2 T/\partial z^2 + q_g/k = (1/\alpha)\,\partial T/\partial t$$

Spherical co-ordinates (r, θ, φ):

$$(1/r^2)\,\partial/\partial r(\,r^2\,\partial T/\partial r\,) + (1/r^2\sin^2\theta)\,\partial^2 T/\partial \varphi^2 + (1/r^2\sin\theta)\,\partial/\partial\theta(\,\sin\theta\,\partial T/\partial\theta\,) + q_g/k = (1/\alpha)\,\partial T/\partial t$$

1-D steady radial forms (no generation): Cylinder: $d/dr(\,r\,dT/dr\,) = 0$ Sphere: $d/dr(\,r^2\,dT/dr\,) = 0$

1.7 Initial and Boundary Conditions

→ Initial condition : $T(x, 0) = T_i$ (needed for unsteady problems)

→ 1st kind (Dirichlet) : surface temperature specified, $T(0,t) = T_s$

→ 2nd kind (Neumann) : heat flux specified, $-k\,\partial T/\partial x\,|_0 = q_0$; insulated surface : $\partial T/\partial x = 0$

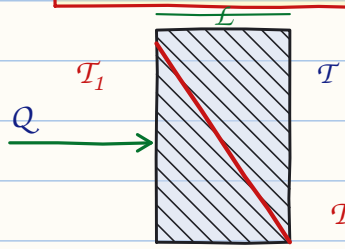
→ 3rd kind (Robin / convection) : $-k\,\partial T/\partial x\,|_0 = h\,[\,T_\infty - T(0)\,]$

Steady Conduction – Plane Wall

Wall thickness L, area A, constant k, no generation, faces at T_1 and T_2. From $d^2T/dx^2 = 0$: $T = C_1x + C_2$. Using $T(0)=T_1$, $T(L)=T_2$:

$$T(x) = T_1 - (T_1 - T_2)\,x / L$$

$$Q = k A\,(T_1 - T_2) / L = (T_1 - T_2) / (L / kA)$$ *linear profile*



T-profile is a straight line

Analogy: $I = \Delta V / R$

$Q = \Delta T / R_{th}$

$R_{th} = L / kA$



→ Thermal resistance : conduction $R = L/kA$ (K/W); convection $R = 1/hA$; radiation $R = 1/h_rA$

Composite (series) wall

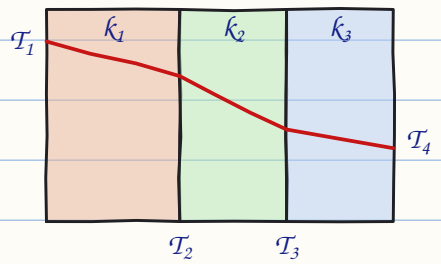
Layers 1,2,3 in series carry the same Q. Add resistances like electrical resistors:

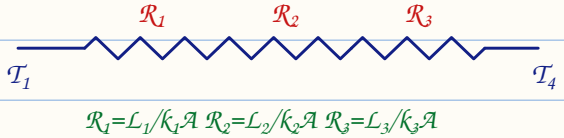
$$Q = (T_1 - T_4) / (\,L_1/k_1A + L_2/k_2A + L_3/k_3A\,)$$

With convection on both sides (hot fluid T_h, cold fluid T_c) :

$$Q = (T_h - T_c) / [\,1/h_1A + \Sigma L/kA + 1/h_2A\,] = U\,A\,(T_h - T_c)$$

$$U = 1 / (\,1/h_1 + \Sigma L/k + 1/h_2\,)\ [W/m^2K]$$ *overall U*





$R_1 = L_1/k_1A$ $R_2 = L_2/k_2A$ $R_3 = L_3/k_3A$

Interface temperatures : $T_2 = T_1 - Q R_1$, $T_3 = T_2 - Q R_2$. Parallel layers : $1/R_{eq} = 1/R_a +$

$1/R_t$ (same ΔT, Q adds).

> **Note:** Contact resistance : imperfect contact between two solids creates a temperature drop ΔT_c at the interface, $R_c = \Delta T_c/Q$ (m^2K/W). It is reduced by increasing pressure or using thermal grease / soft foil.

Solved Example 1 – Furnace wall

Wall: fire-brick L=0.25 m, k=1.0; insulation L=0.10 m, k=0.05; red brick L=0.20 m, k=0.8 W/mK. Inner face 1000°C, outer face 50°C. Find heat loss per m^2 and interface temperatures.

$R/A = 0.25/1 + 0.10/0.05 + 0.20/0.8 = 0.25 + 2.0 + 0.25 =$ **2.5 m^2K/W**

$q = (1000-50)/2.5 =$ **380 W/m^2**; $T_2 = 1000 - 380(0.25) =$ **905°C**; $T_3 = 905 - 380(2.0) =$ **145°C**

Plane wall with internal heat generation

Slab of thickness $2L$, both faces held at T_w, uniform generation q_g (W/m^3). From $d^2T/dx^2 + q_g/k = 0$ with symmetry ($dT/dx = 0$ at $x = 0$):

$$T(x) = T_w + (q_g/2k)(L^2 - x^2)$$

$$T_{max} = T_w + q_gL^2/2k \text{ (at the mid-plane)} \qquad \text{parabolic}$$

→ Long solid cylinder (radius R): $T = T_w + (q_gR^2/4k)(1 - r^2/R^2)$, $T_{max} = T_w + q_gR^2/4k$

→ Solid sphere: $T_{max} = T_w + q_gR^2/6k$. Heat generated per unit volume = heat lost at the surface (steady).

Solved Example 1B – Wall with convection on both sides

Brick wall L = 0.3 m, k = 0.6 W/mK. Inside air 25°C (h_i = 8), outside air -5°C (h_o = 20 W/m^2K). Find U, heat loss and inside surface temperature.

$R = 1/8 + 0.3/0.6 + 1/20 = 0.125 + 0.5 + 0.05 = 0.675\ m^2K/W \rightarrow$ **$U = 1.48\ W/m^2K$**

$q = 30/0.675 =$ **44.4 W/m^2**; $T_{s,in} = 25 - 44.4(0.125) =$ **19.4°C**

Steady Conduction – Cylinders & Spheres

Hollow cylinder: inner radius r_1 (T_1), outer radius r_2 (T_2), length L. Radial conduction, no generation: $d/dr(\ r\ dT/dr\ ) = 0 \rightarrow T = C_1 \ln r + C_2$. Heat flow through any radius is constant, $Q = -k(2\pi rL)\ dT/dr$. Integrating from r_1 to r_2:

$$Q = 2\pi k L (T_1 - T_2) / \ln(r_2/r_1)$$

$$R_{cyl} = \ln(r_2/r_1) / (2\pi k L)$$ cylinder

→ Temperature profile is logarithmic (not linear): $(T - T_1)/(T_2 - T_1) = \ln(r/r_1) / \ln(r_2/r_1)$

→ Heat flux $q = Q/A$ decreases with r (area $2\pi rL$ increases) but Q stays constant

Composite cylinder (pipe + insulation) with convection inside (h_i) and outside (h_o):

$$Q = (T_i - T_o) / [\ 1/(h_i A_i) + \ln(r_2/r_1)/(2\pi k_1 L) + \ln(r_3/r_2)/(2\pi k_2 L) + 1/(h_o A_o)\ ]$$

Hollow sphere: $d/dr(\ r^2\ dT/dr\ ) = 0 \rightarrow T$ varies as $1/r$:

$$Q = 4\pi k (T_1 - T_2) / (\ 1/r_1 - 1/r_2\ )$$

$$R_{sph} = (r_2 - r_1) / (4\pi k r_1 r_2)$$ sphere

Solved Example 2 – Insulated steam pipe

Steam pipe, outer radius r_1=50 mm at 200°C, covered with 50 mm insulation (k=0.1 W/mK); outer surface of insulation at 40°C. Heat loss per metre?

r_2 = 100 mm. $Q/L = 2\pi k (T_1-T_2) / \ln(r_2/r_1) = 2\pi(0.1)(160) / \ln 2 = 100.5/0.693 =$ 145 W/m

Solved Example 2B – Pipe with insulation and convection

Pipe surface 250°C, r_1 = 30 mm, covered with 25 mm of insulation (k = 0.08 W/mK). Ambient air 30°C, h_o = 12 W/m²K. Find the loss per metre and the outer surface temperature of the insulation.

r_2 = 55 mm; $R_{ins} = \ln(55/30) / (2\pi \times 0.08) = 0.606/0.503 = 1.206$ mK/W

$R_{conv} = 1/(2\pi\ r_2\ h_o) = 1/(2\pi \times 0.055 \times 12) = 0.241$ mK/W

$Q/L = (250 - 30)/(1.206 + 0.241) =$ 152 W/m; $T_{surf} = 30 + 152(0.241) =$ 66.7°C

Critical Thickness (Radius) of Insulation

Adding insulation to a small pipe increases the conduction resistance but also increases the outer surface area (lower convection resistance). So heat loss may first rise and then fall.

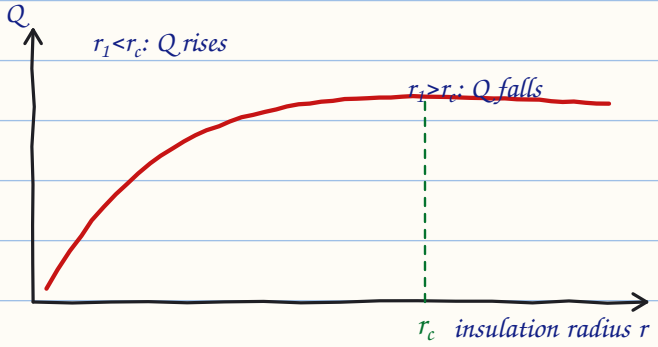
Pipe radius r_1, insulation up to radius r, conductivity k, outside convection h_o:

$$R_{tot} = \ln(r/r_1) / (2\pi k L) + 1 / (2\pi\ r\ L\ h_o)$$

For maximum heat loss (minimum R_{tot}) put $dR_{tot}/dr = 0$: $1/(2\pi kLr) - 1/(2\pi\ r^2\ L\ h_o) = 0$

$$r_c = k / h_o \text{ (cylinder)}$$
$$r_c = 2k / h_o \text{ (sphere)}$$
critical radius

$d^2R/dr^2 > 0$ at $r = r_c$, so R is a *minimum* and Q a *maximum* there.



$r_1 < r_c$: insulation *raises* loss

$r_1 > r_c$: insulation *reduces* loss

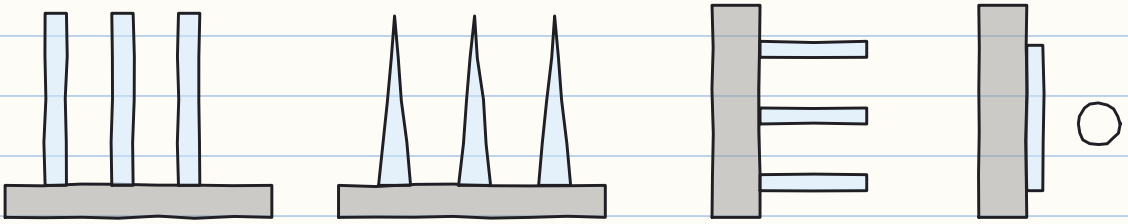
Insulation effective only if $r_1 \geq r_c$

→ Critical thickness = $r_c - r_1$. Used in electric wires (insulation cools them, $r_1<r_c$) and small tubes.

Note: For steam pipes r_1 is large ($r_1 > k/h \approx 20–50$ mm) so any insulation reduces the loss. Example: $k=0.5$, $h=10 \rightarrow r_c = 0.05$ m = 50 mm.

Heat Conduction through Fins

A *fin* (extended surface) is a protrusion that increases the heat dissipation area: $Q = hA(T_s-T_\infty)$ rises when A rises. Used on IC-engine cylinders, radiators, electronic heat sinks, condensers, transformers.



Assumptions & governing equation

→ 1-D steady conduction along fin; constant k, h, T_∞; no generation; negligible radiation; fin thin (Biot small)

Energy balance on element dx: $Q_x = Q_{x+dx} + h\,P\,dx\,(T - T_\infty)$. Put $\theta = T - T_\infty$, P = perimeter, A_c = cross-section area:

$$d^2\theta/dx^2 - m^2\,\theta = 0\,,\; m = \sqrt{(hP / kA_c)}$$
$$\theta = C_1\, e^{mx} + C_2\, e^{-mx}$$
fin equation

Base condition $\theta(0) = \theta_b = T_b - T_\infty$. Tip condition gives three cases (fin length L):

Tip condition	θ/θ_b	Heat dissipated Q
Very long fin	e^{-mx}	$\sqrt{(hPkA_c)}\,\theta_b$
Insulated tip	$\cosh m(L-x) / \cosh mL$	$\sqrt{(hPkA_c)}\,\theta_b \tanh mL$
Convecting tip	see formulas below	see formulas below

$$\theta/\theta_b = [\cosh m(L-x) + (h/mk)\sinh m(L-x)] / [\cosh mL + (h/mk)\sinh mL]$$

$$Q = \sqrt{(hPkA_c)}\,\theta_b\,[\sinh mL + (h/mk)\cosh mL] / [\cosh mL + (h/mk)\sinh mL]$$

Note: Convecting tip is handled easily with corrected length in the insulated-tip formula : $L_c = L + t/2$ (rectangular fin, thickness t), $L_c = L + D/4$ (pin fin).

Fin efficiency & effectiveness

$$\eta_{fin} = Q_{fin} / Q_{max} = Q_{fin} / (h A_{fin} \theta_b)$$ (insulated tip : $\eta = \tanh mL / mL$)

$$\varepsilon_{fin} = Q_{fin} / (h A_c \theta_b)$$ (very long fin : $\varepsilon = \sqrt{(kP / hA_c)}$) — η and ε

→ Q_{max} = heat if the whole fin were at base temperature. $\eta \to 1$ as $mL \to 0$; η drops for long fins

→ ε is the gain from adding the fin: $\varepsilon = \eta\,(A_{fin}/A_c)$. Fin is worthwhile only if $\varepsilon > 2$

→ To raise ε: high k material (Cu, Al), thin closely spaced fins, fins on the side with low h (gas side), large P/A_c

Solved Example 3 – Pin fin (insulated tip)

Al rod D=10 mm, L=100 mm, k=200 W/mK, h=20 W/m²K, T_b=120°C, T_∞=30°C.

$m = \sqrt{(4h/kD)} = \sqrt{(4\times20/(200\times0.01))} = \sqrt{40} = 6.325\ m^{-1}$; $mL = 0.6325$; $\tanh mL = 0.560$

$\sqrt{(hPkA_c)} = \sqrt{(20\times0.0314\times200\times7.854\times10^{-5})} = 0.0993\ W/K$

$Q = 0.0993 \times 90 \times 0.560 = 5.0\ W$; $Q_{max} = hPL\theta_b = 5.65\ W$; $\eta = 0.885$

Solved Example 3B – Rectangular fin (convecting tip)

Aluminium fin, thickness t = 3 mm, length L = 40 mm, width w = 1 m, k = 200, h = 40 W/m²K, T_b = 100°C, T_∞ = 30°C. Use corrected length.

$L_c = L + t/2 = 41.5$ mm; $P \approx 2w = 2.006$ m; $A_c = wt = 0.003\ m^2$; $m = \sqrt{(hP/kA_c)} = \sqrt{133.7} = 11.56\ m^{-1}$

$mL_c = 0.480$; $\tanh mL_c = 0.446$; $\sqrt{(hPkA_c)} = \sqrt{(40\times2.006\times200\times0.003)} = 6.94\ W/K$

$Q = 6.94 \times 70 \times 0.446 = 217\ W$; $\eta = \tanh mL_c / mL_c = 0.93$; $\varepsilon = Q/(hA_c\theta_b) = 217/(40\times0.003\times70) = 25.8$

Use of fins – temperature measurement

A thermometer well (hollow tube) in a pipe behaves like a fin with insulated tip. The sensing tip reads $T_L = T_\infty + \theta_b/\cosh mL$, so error = $(T_L - T_\infty)$ is minimised by large mL: long, thin, low-k well and high h.

→ Hence: error ∝ $1/\cosh mL$ → use long well, thin walls, material of low k (stainless steel), insulate the outside of pipe

→ Fin array (n fins): $Q_{total} = n\, Q_{fin} + h\, A_{unfinned}\, \theta_b$; overall surface effectiveness $\eta_o = 1 - (n\, A_f/A_t)(1 - \eta_f)$

Note: Fin design rule: fins are useful when h is small (gases, natural convection). On a water side (h large) fins give little benefit; and ε is the figure of merit while η tells about material utilisation.

Unsteady (Transient) Heat Conduction

Temperature varies with time as well as position: $\partial T/\partial t \neq 0$. Occurs in heating/cooling of billets, quenching, start-up and shut-down of equipment, thermocouple response.

Lumped System Analysis

Assume the body is at a uniform temperature at every instant (internal conduction resistance negligible compared with surface convection). Valid when the Biot number is small:

$$Bi = h\, L_c / k = (\text{internal resistance})/(\text{external resistance}) < 0.1\ ,\ L_c = V/A_s \quad \text{criterion}$$

→ L_c: sphere R/3, long cylinder R/2, cube a/6, plate of thickness 2L → L (both faces exposed)

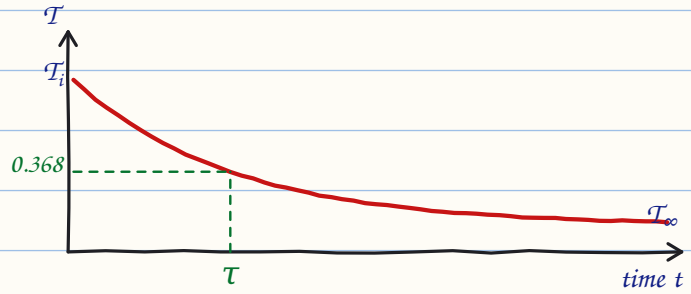
Body (ρ, V, c, T_i) suddenly placed in fluid at T_∞. Energy balance: $-hA_s(T - T_\infty) = \rho\, V\, c\, dT/dt$. With $\theta = T - T_\infty$ and $\theta_i = T_i - T_\infty$:

$$\theta/\theta_i = (T - T_\infty)/(T_i - T_\infty) = \exp(-hA_s t/\rho V c) = \exp(-Bi \cdot Fo) \quad \text{lumped solution}$$

→ $Fo = \alpha\, t / L_c^2$ (Fourier number, dimensionless time)

→ Time constant $\tau = \rho V c / hA_s = R_{conv} \cdot C_{thermal}$; at $t = \tau$, $\theta/\theta_i = e^{-1} = 0.368$ (63.2% of change done)

→ Total heat transferred in time t : $Q = \rho\, V\, c\, (T_i - T_\infty)\, [\, 1 - e^{-t/\tau}\,]$



Exponential decay of θ

Small τ → fast response

(small mass / large h)

Solved Example 4 – Cooling of a steel ball

Ball D=10 mm, ρ=7800 kg/m³, c=470 J/kgK, k=45; cools from 400°C in air at 30°C with h=50 W/m²K to 100°C. Find time.

$L_c = D/6 = 1.667\times10^{-3}$ m; $Bi = 50(1.667\times10^{-3})/45 = 0.0019 < 0.1$ → lumped OK

$\tau = \rho\, c\, L_c/h = 7800\times470\times1.667\times10^{-3}/50 = 122$ s; $(100-30)/(400-30) = 0.189 = e^{-t/\tau}$

$t = \tau \ln(1/0.189) = 122 \times 1.665 =$ **203 s**

Heisler (Transient-Temperature) Charts – Bi > 0.1

When Bi > 0.1, temperature varies inside the body. For a plane wall (thickness 2L), long cylinder or sphere (radius r_o) with uniform T_i suddenly exposed to fluid T_∞ (convection), the exact series solution is plotted as Heisler charts using:

$$Bi = h\,L\,/\,k \text{ (or } 1/Bi = k/hL)$$

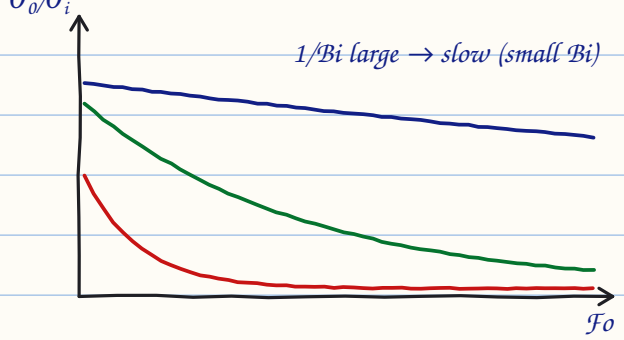
$$Fo = \alpha\, t\,/\,L^2 \text{ (L = half thickness, or } r_o)$$

$$\theta^* = (T - T_\infty)/(T_i - T_\infty)$$

→ Chart 1 : centre-line temperature θ_0/θ_i vs Fo for various 1/Bi (x = 0 or r = 0)

→ Chart 2 : position correction θ/θ_0 vs 1/Bi for various x/L (or r/r_o); independent of time

→ Chart 3 : heat loss ratio Q/Q_0 vs Bi^2Fo for different Bi; $Q_0 = \rho V c(T_i - T_\infty)$



Centre temp. $\theta_0 = T_0 - T_\infty$

Then $\theta = \theta_0 \times$ (chart 2)

Q/Q_0 from chart 3

One-term fit (Fo > 0.2):

$\theta_0/\theta_i = C_1\, e^{-\zeta_1^2 Fo}$

One-term approximation (Fo > 0.2) for a plane wall: $\theta_0/\theta_i = C_1 \exp(-\zeta_1^2 Fo)$, $\zeta_1 \tan\zeta_1 = Bi$, $C_1 = 4\sin\zeta_1/(2\zeta_1 + \sin 2\zeta_1)$. Example: Bi = 1 → ζ_1=0.860, C_1=1.119; for Fo = 0.5, $\theta_0/\theta_i = 1.119\, e^{-0.37} =$ **0.773**.

Procedure for a Heisler-chart problem

1. Compute $Bi = hL/k$. If $Bi < 0.1$ use the lumped method, else Heisler.
2. $Fo = \alpha t/L^2$; read θ_0/θ_i from chart 1 → get T_0.
3. For a point at x/L, read θ/θ_0 from chart 2; $T = T_\infty + (\theta/\theta_0)(\theta_0/\theta_i)\theta_i$.
4. For total heat, use $Bi^2 Fo$ in chart 3 → $Q = Q_0(Q/Q_0)$.

Extra: Semi-infinite solid (thick body, early times): for surface at T_s suddenly imposed, $(T - T_s)/(T_i - T_s) = erf(x / 2\sqrt{(\alpha t)})$. $Fo < 0.2$ → use this, not the one-term formula.

Multi-dimensional transient – product solution

Temperature in a short cylinder, rectangular bar or semi-infinite plate is the product of 1-D solutions:

$$\theta^*_{short\ cyl}(r,x,t) = \theta^*_{plane\ wall}(x,t) \times \theta^*_{infinite\ cyl}(r,t)$$

$$\theta^*_{bar}(x,y,t) = \theta^*_{slab\ 1}(x,t) \times \theta^*_{slab\ 2}(y,t)$$ Newman product rule

→ Each 1-D factor is read from the Heisler chart using its own Bi and Fo (own half-thickness / radius)

Significance of dimensionless numbers

Number	Definition	Meaning
Biot Bi	hL_c/k	internal / surface resistance
Fourier Fo	$\alpha t/L_c^2$	conduction rate / storage rate
Thermal time constant τ	$\rho Vc/hA$	time for 63.2% response

Solved Example 5 – Is lumped analysis valid?

Aluminium cube, side 5 cm ($k = 205$, $\rho = 2700$, $c = 900$), dropped in water with $h = 500$ W/m^2K.

$L_c = a/6 = 8.33$ mm; $Bi = 500(0.00833)/205 = 0.0203 < 0.1$ → lumped valid. $\tau = \rho c L_c/h = 2700\times900\times0.00833/500 = 40.5$ s

For the same cube with $h = 5000$: $Bi = 0.203 > 0.1$ → use Heisler charts.

Solved Example 6 – Heisler procedure (plane wall)

Large steel plate $2L = 10$ cm ($k = 45$, $\alpha = 1.2\times10^{-5}$ m^2/s) at 500°C quenched in oil at 50°C, $h = 450$ W/m^2K. Find the mid-plane temperature after 3 min.

$L = 0.05$ m; $Bi = hL/k = 450(0.05)/45 = 0.5$; $Fo = \alpha t/L^2 = 1.2\times10^{-5}(180)/0.0025 = 0.864$ (> 0.2, one-term OK)

For $Bi = 0.5$: $\zeta_1 = 0.6533$, $C_1 = 1.0701$. $\theta_0/\theta_i = 1.0701\ exp(-0.4268 \times 0.864) =$

$1.0701 \times 0.6915 = 0.740$

$T_0 = 50 + 0.740(500 - 50) = 383°C$

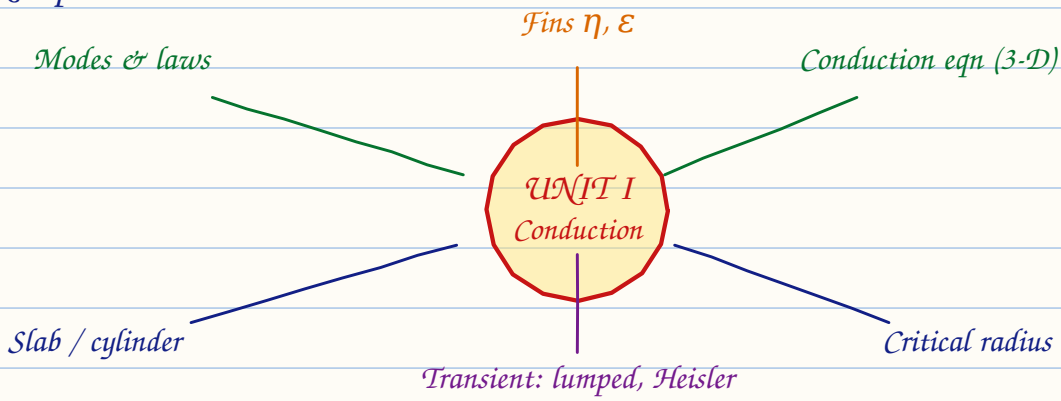
Important Questions (exam oriented)

Short answer (2 marks):

1. Define the three modes of heat transfer. State Fourier's law.
2. What is thermal diffusivity? Why is it important in transient conduction?
3. Define critical radius of insulation. Give its expression for a cylinder and a sphere.
4. Define fin efficiency and fin effectiveness. When is a fin justified?
5. What is the Biot number? State the condition for the lumped-capacity method.

Long answer (10 marks):

1. Derive the general 3-D heat conduction equation in Cartesian co-ordinates. Reduce it to Fourier, Poisson and Laplace equations.
2. Derive an expression for the heat flow through a composite cylinder with convection on both sides.
3. Derive the critical radius of insulation for a cylinder. Explain its significance.
4. Derive the temperature distribution and heat dissipation of a fin with insulated tip.
5. Derive the lumped-system temperature-time relation. Explain Heisler charts with an example.



Quick Revision – Formulae to remember

Item	Formula
Plane wall	$Q = kA\Delta T/L$; $R = L/kA$
Cylinder	$Q = 2\pi kL\Delta T / \ln(r_2/r_1)$
Sphere	$Q = 4\pi k\Delta T / (1/r_1 - 1/r_2)$
Critical radius	cyl: k/h_o sph: $2k/h_o$
Fin (insulated tip)	$Q = \sqrt{(hPkA)}\,\theta_b \tanh mL$
Lumped system	$\theta/\theta_i = e^{-t/\tau}$, $Bi < 0.1$